

Web Services/API Document

for

Integration with Financial Institution

Submitted to



**WAREHOUSE
RECEIPT SYSTEM
COUNCIL**

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1. INTRODUCTION

E-warehouse is a platform for managing warehouse operations, depositor registrations, loan sanctions, and auction processes. The APIs facilitate secure data transactions and ensure seamless integration with external systems.

2. PURPOSE OF THIS DOCUMENT

This This document serves as a reference guide for stakeholders to understand the API endpoints, their usage, and integration with the e-Warehouse system.

3. HOW TO ACCESS THE API SERVICES

- ❖ **Registration of Application & Whitelisting of IP Address**
- ❖ **Token-Based Authentication using OAuth2**
- ❖ **Secure Communication via HTTPS**

4. LIST OF SERVICES

4.1 API for: Get OAuth Token.

❖ **Use Cases:**

This API will be helpful to get the Token no. for authenticating the user. Stakeholder has to provide the specified User Name and Password to get the Token No. This token no will be valid for 30 minutes and expires automatically. This token no. to be passed in header in every APIs.

❖ **Service Request Name and URL**

URL	http://localhost:8888/oauth2/token
Method	POST

❖ **Authentication Details**

- **Auth Type:** Basic Auth
- **Username:** winnie gwada
- **Password:** winnie123

❖ **Request Header**

Header	Value
Authorization	*****
Cache-Control	no-cache
Postman-Token	<calculated when request is sent>
Content-Type	application/x-www-form-urlencoded
Content-Length	<calculated when request is sent>
Host	<calculated when request is sent>
User-Agent	PostmanRuntime/7.43.3
Accept	/
Accept-Encoding	gzip, deflate, br
Connection	keep-alive

❖ **Request Body (Form-Data/ x-www-form-urlencoded)**

Key	Value
grant_type	client_credentials

client_id	winnie_gwada
client_secret	winnie123
scope	read write

❖ **Sample Request (cURL Format):**

```
http://localhost:8888/api/
eyJhbGciOiJSUzI1NiJ9.eyJzdWUiOiJwcml5YW5rYSIsImF1ZCI6ImByaXlhbmthliwibmJmIjoxNzQzNDkwNzQ5LCJzY29wZSI6WyJyZWFKliwid3JpdGUiXSwiaXNzIjoiaHR0cDovL2xvY2FsaG9zdDo4ODg4IiwiaXhwaWJjoxNzQzNDk0MzQ5LCJpYXQiOiJlE3NDM0OTA3NDksImp0aSI6IjU5MTkyZjRjLWI5OGUtNDJkOC1hYWU0LWFjZmMxODgxNDA2MCI9.gAf-rFXyIM2uJc4O-GFtazSSyZZEDHGFaLw_isNxy-_Ut_hau8MK7-6nguH5muVPm1OPU1LkKQwFQSBXghxyxrT4N-SkIVepAG4o_3N9Ro8sZ7hs8atyEM_38FxTD6Eb42tFfYcpM2-_j572mj_ibKPTy5SrnITXchK2TSwc_61IQSxESfyQsfD_i3iazE-sFDS3czrJ0RsKnDpHir_FAESfutn_Y7eoPUtFJgH1SivtnR4PD7PgKvklfMr31w2MdGSPimKxpHQLtTHglu_TIElcfQexx7EP9Y0wO41UcQE1t7WBpn6WqEoEyu_2dpC1IBB6KOWliP7oT-r8Tox9w
```

❖ **Sample Response Template:**

```
{
  "access_token": "
eyJhbGciOiJSUzI1NiJ9.eyJzdWUiOiJwcml5YW5rYSIsImF1ZCI6ImByaXlhbmthliwibmJmIjoxNzQzNDkwNzQ5LCJzY29wZSI6WyJyZWFKliwid3JpdGUiXSwiaXNzIjoiaHR0cDovL2xvY2FsaG9zdDo4ODg4IiwiaXhwaWJjoxNzQzNDk0MzQ5LCJpYXQiOiJlE3NDM0OTA3NDksImp0aSI6IjU5MTkyZjRjLWI5OGUtNDJkOC1hYWU0LWFjZmMxODgxNDA2MCI9.gAf-rFXyIM2uJc4O-GFtazSSyZZEDHGFaLw_isNxy-_Ut_hau8MK7-6nguH5muVPm1OPU1LkKQwFQSBXghxyxrT4N-SkIVepAG4o_3N9Ro8sZ7hs8atyEM_38FxTD6Eb42tFfYcpM2-_j572mj_ibKPTy5SrnITXchK2TSwc_61IQSxESfyQsfD_i3iazE-sFDS3czrJ0RsKnDpHir_FAESfutn_Y7eoPUtFJgH1SivtnR4PD7PgKvklfMr31w2MdGSPimKxpHQLtTHglu_TIElcfQexx7EP9Y0wO41UcQE1t7WBpn6WqEoEyu_2dpC1IBB6KOWliP7oT-r8Tox9w",
  "token_type": "bearer",
  "expires_in": 3600,
  "scope": "read write"
}
```

4.2 API for: Fetching Financial Institution

❖ **Use Cases**

This API retrieves the list of all financial institutions available in the system. It is primarily used by authorized users to view and reference financial institution details for

informational or validation purposes. Upon successful request, the complete list of institutions is returned

❖ **Service Request Name and URL**

URL	http://localhost:8888/api/finance/getFinancialInstitution
Method	GET

❖ **Header**

Token

```
eyJhbGciOiJIUzI1NiJ9.eyJzdWUiOiJwcmI5YW5rYSIsImF1ZCI6ImByaXlhbGthliwibmJmIjoxNzQzNDk5OTE4LCJzY29wZSI6WyJyZWZkIiwid3JpdGUiXSwiaXNzIjoiaHR0cDovL2xvY2FsaG9zdDo4ODg4IiwiaXhwaWJjoxNzQzNTAzNTE4LCJpYXQiOiJlE3NDM0OTk5MTg5IiwiaWF0Ij0iMj01MDQzZDFjLWV3NGI0ZWYyMTRhYyJ9.AwhcN6tVutxY7M1YLyQe6QtKUZZlepB29x5wiOETljkj390w51RkpxlqJc8Z4uvRp0TJzxo4xOXDALCLzUus4eogHpmkq66HfxnQ-LDhKq1bnw-lgMtrvCJWrjwNiKCwh9F0LpzDkYPkV1UAlaWVPBjP97XO-zNlvTnkgomsIfkGgaTGab5bfXwukjgURf3zi1KW2JJKGNC5GXf72zYWB-PBSdjtH4J6YPCdSt4s9p3uDVBemV1WSq1ZnUeghYhEKMN-m32k3VYiso8amfJE_ndaij8lHEOIBNXpq3dRJkiUFLfzfdhGVAcqdto6Wdxw-iWo44qSOPBueDV-Ee0rA
```

❖ **Sample Response Template**

```
{
  "financialInstitutionId": "1",
  "financialInstitutionName": "Bank Of Africa"
},
{
  "financialInstitutionId": "2",
  "financialInstitutionName": "Equity Bank"
},
{
  "financialInstitutionId": "3",
  "financialInstitutionName": "KCB Bank"
}
```

Status Codes:

- 200 - OK
- 400 - Bad Request
- 401– Unauthorized (Invalid token)
- 404– Not Found (No data for provided ID)
- 500– Internal Server Error (Third-party service failure)

4.3 API for: Pledging of Warehouse Receipt

❖ Use Cases

This API is used to initiate a pledge request against a warehouse receipt by its unique receipt ID. It is primarily intended for depositors or authorized users who wish to use a registered warehouse receipt as collateral. Upon submission, the pledge request will be sent to the financial institution, and after reviewing the application, the institution will provide an appropriate response to the user. The API ensures that only eligible receipts (e.g., registered, not already pledged or retired) can be used for pledge applications.

To assist in the decision-making process, the financial institution is expected to verify the receipt details at the time of approval by calling the API:

`http://localhost:8888/api/getIssuanceDetailsById/{issuanceId}`

❖ Service Request Name and URL

URL	<code>http://localhost:8888/api/loans/savePledging</code>
Method	POST

❖ Header

Token

`eyJhbGciOiJSUzI1NiJ9.eyJzdWUiOiJwcmI5YW5rYSIsImF1ZCI6ImByaXlhbWthliwibmJmIjoxNzQzNDk5OTE4LCJzY29wZSI6WyJyZWFKliwid3JpdGUiXSwiaXNzIjoiaHR0cDovL2xvY2FsaG9zdDo4ODg4IiwiaXhwaWljoxNzQzNTAzNTE4LCJpYXQiOiJlE3NDM0OTk5MTgsImp0aSI6ImVIZDkxZjQyLWlyMDQtNGJlYy04ZDFjLWw3NGI0ZWYyMTRhYyJ9.AwhcN6tVutxY7M1YLyQe6QtKUZZlep29x5wiOETljkjJ390w51RkpxlqJc8Z4uvRp0TJzxo4xOXDALCLzUus4eogHpmkq66HfxnQ-LDhKq1bnw-IgMtrvCJWrjwNiKCwh9F0LpzDkYPkV1UAlaWVPBjP97XO-zNlvTnkgomslfkGgaTGab5bfXwukjgURf3zi1KW2JJKGNC5GXf72zYWB-PBSdjtH4J6YPCdSt4s9p3uDVBemV1WSq1ZnUeghYhEKMNm32k3VYiso8amfJE_ndaij8lHEOlBNXpq3dRjkiUFLfzfdhGVAcqdto6Wdxw-iWo44qSOPBueDV-Ee0rA`

❖ Sample Request Template

```
{
  "financialInstitutionId": "1",
  "dateOfPledging": "20 May, 2025",
  "proposedLoanAmount": 95000,
  "proposedDateOfLoanClearance": "12 March, 2026",
  "durationOfLoan": 9,
  "depositorId": "8J733J3H57",
  "issueld": 7,
  "userId": 204,
```

```
"userName": "Alice",
"roleId": 6,
"roleName": "Warehouse Operator"
}
```

❖ *Sample Response Template*

```
{
  "status": "success",
  "message": "Pledging applied successfully"
}
```

Status Codes:

- 200 - OK
- 400 - Bad Request
- 401– Unauthorized (Invalid token)
- 404– Not Found (No data for provided ID)
- 500– Internal Server Error (Third-party service failure)

4.4 API for: Loan Details

◆ Use Cases

This API fetches the loan sanction status and related details associated with a specific warehouse receipt or issuance based on the provided `issuanceId` . It is primarily used by authorized users such as financial institution officers and administrators to determine whether a loan has been sanctioned or rejected. Upon successful request, the API returns the loan status along with relevant sanction or rejection details for validation, processing, or reference.

❖ *Service Request Name and URL*

URL	http://localhost:8888/api/loan-details
Method	GET

◆ *Header*

Token

eyJhbGciOiJIUzI1NiJ9.eyJzdWUiOiJwcmI5YW5rYSIsImF1ZCI6InByaXlhbmthliwibmJmIjoxNzQzNDk5OTE4LCJzY29wZSI6WyJyZWZkIiwid3JpdGUiXSwiaXNzIjoiaHR0cDovL2xvY2FsaG9zdDo4ODg4IiwiZXhwIjoxNzQzNTAzNTE4LCJpYXQiOiJlE3NDM0OTk5MTgsImp0aSI6ImVIZDkxZjQyLWlyMDQtNGJlYy04ZDFjLWw3NGI0ZWYyMTRhYyJ9.AwhcN6tVutxY7M1YLyQe6QtKUZZlepb29x5wiOETljkjJ390w51RkpxlqJc8Z4uvRp0TJzxo4xOXDALCLzUus4eogHpmkq66HfxnQ-LDhKq1bnw-lgMtrvCJWrjwNiKCwh9F0LpzDkYPkV1UAlaWVPBjP97XO-zNlvTnkgomsIfkGgaTGab5bfXwukjgURf3zi1KW2JJKGNC5GXf72zYWB-PBSdjth4J6YPCdSt4s9p3uDVBemV1WSq1ZnUeghYhEKMN-m32k3VYiso8amfJE ndaij8lHEOIBNXpq3dRJkiUFLfzfdhGVAcgdto6Wdxw-

iWo44qSOPBueDV-Ee0rA

❖ **Sample Request Template**

```
{
  "wareHouseReciptNo": "WR-124-2025",
  "issuanceWhld": 7
}
```

❖ **Sample Response Template**

```
{
  "financialInstitution": 1,
  "dateOfPledging": "2025-05-20",
  "proposedLoanAmount": 95000,
  "proposedDateOfLoanClearance": "2026-03-12",
  "durationOfLoan": 9
}
```

Status Codes:

- 200 - OK
- 400 - Bad Request
- 401– Unauthorized (Invalid token)
- 404– Not Found (No data for provided ID)
- 500– Internal Server Error (Third-party service failure)

4.5 API for: Loan Sanction

❖ **Use Cases**

This API fetches the loan sanction status and related details associated with a specific warehouse receipt or issuance based on the provided issuanceId . It is primarily used by authorized users such as financial institution officers and administrators to determine whether a loan has been sanctioned or rejected. Upon successful request, the API returns the loan status along with relevant sanction or rejection details for validation, processing, or reference.

❖ **Service Request Name and URL**

URL	http://localhost:8888/api/sanction-details
Method	GET

❖ **Header**

Token

```
eyJhbGciOiJSUzI1NiJ9.eyJzdWUiOiJwcmI5YW5rYSIsImF1ZCI6InByaXlhbmthliwibmJmIjoxNzQzNDk5OTE4LCJzY29wZSI6WyJyZWFKliwid3JpdGUiXSwiaXNzIjoiaHR0cDovL2xvY2FsaG9zdDo4ODg4IiwiaXhwIjoxNzQzNTAzNTE4LCJpYXQiOiJlE3NDM0OTk5MTgslmp0aSI6ImVIZDkxZjQyLWlyMDQtNGJlYy04ZDFjLWM3NGI0ZWYyMTRhYyJ9.AwhcN6tVutxY7M1YLyQe6QtKUZZlepb29x5wiOETljkjJ390w51RkpxlqJc8Z4uvRp0TJzxo4xOXDALCLzUus4eogHpmkq66HfxnQ-LDhKq1bnw-IgMtrvCJWrjwNiKCwh9F0LpzDkYPkV1UAlaWVPBjP97XO-zNlvTnkgomslfkGgaTGab5bfXwukjgURf3zi1KW2JJKGNC5GXf72zYWB-PBSdjtH4J6YPCdSt4s9p3uDVBemV1WSq1ZnUeghYhEKMN-m32k3VYiso8amfJE_ndaij8lHEOlBNXpq3dRJkiUFLfzfdhGVAcqdto6Wdxw-iWo44qSOPBueDV-Ee0rA
```

❖ **Sample Request Template**

```
{
  "wareHouseReceiptNo": "WR-124-2025",
  "issuanceWhId": 7,
  "financialLoanId": 1
}
```

❖ **Sample Response Template**

```
{
  "sanctionedLoanAmount": 85500,
  "approvedDateOfLoanClearance": "2026-06-12",
  "approvedDurationOfLoan": 12,
  "interestRate": 8,
  "loanStatus": "SANCTIONED",
  "rejectionReason": null,
}
```

Status Codes:

- 200 - OK
- 400 - Bad Request
- 401– Unauthorized (Invalid token)
- 404– Not Found (No data for provided ID)
- 500– Internal Server Error (Third-party service failure)

4.6 API for: Fetching Issuance details against warehouse Receipt

❖ **Use Cases**

This API fetches issuance warehouse receipt details based on the provided warehouse Receipt. It is primarily used by authorized users such as warehouse operators and administrators to retrieve detailed information related to a specific warehouse receipt. Upon successful request, the corresponding receipt details are returned from the system for validation or reference.

❖ **Service Request Name and URL**

URL	http://localhost:8888/api/getIssuanceDetailsById
Method	GET

❖ **Header****Token**

eyJhbGciOiJSUzI1NiJ9.eyJzdWUiOiJwcmI5YW5rYSIsImF1ZCI6InByaXlhbmthliwibmJmljoxNzQzNDk5OTE4LCJzY29wZSI6WyJyZWFKliwid3JpdGUiXSwiaXNzIjoiaHR0cDovL2xvY2FsaG9zdDo4ODg4IiwiaXhwaWJjoxNzQzNTAzNTE4LCJpYXQiOiE3NDM0OTk5MTgsImp0aSI6ImVIZDkxZjQyLWlyMDQtNGJlYy04ZDFjLWM3NGI0ZWYyMTRhYyJ9.AwhcN6tVutxY7M1YLyQe6QtKUZZlep29x5wiOETljkj390w51RkpxlqJc8Z4uvRp0TJzxo4xOXDALCLzUus4eogHpmkq66HfxnQ-LDhKq1bnw-IgMtrvCJWrjwNiKCwh9F0LpzDkYPkV1UAlaWVPBjP97XO-zNlvTnkgomslfkGgaTGab5bfXwukjgURf3zi1KW2JJKGNC5GXf72zYWB-PBSdjtH4J6YPCdSt4s9p3uDVBemV1WSq1ZnUeghYhEKMN-m32k3VYiso8amfJE_ndaij8lHEOlBNXpq3dRjkiUFLfzfdhGVAcqdto6Wdxw-iWo44qSOPBueDV-Ee0rA

❖ **Sample Request Template**

```
{
  "wareHouseReciptNo": "WR-124-2025",
}
```

❖ **Sample Response Template**

```
{
  "issuanceWhId": 7,
  "intDepositorWhOperator": {
    "intDepositorWhOperator": "8J733J3H57",
    "typeOfUser": "Depositor",
    "wareHouseName": {
      "vchWarehouselD": "WAR20250227001",
      "vchCompanyName": "Gyana Company",
      "vchWarehouseName": "Warehouse One",
    },
    "companyName": "COP100041",
    "typeOfDepositor": "Individual",
    "nameOfDepositor": "Paresh Mohanta",
    "mobileNumber": "+254888765456",
    "emailAddress": "pareshMohantaOne@gmail.com",
    "postalCode": "44444",
    "postalAddress": "kenya",
    "countyId": {
      "id": 56,
    }
  }
}
```

```
"name": "Nairobi1",
"subCounties": [
  {
    "intId": 23,
    "txtSubCountyName": "Westlands",
    "wards": [
      {
        "intId": 1,
        "vchWardName": "Kangemi",
      }
    ]
  },
]
},
"intNationality": {
  "countryId": 34,
  "countryName": "Kenya",
  "countryCode": "KEN"
},
"entityRegistrationNumber": "",
"depositorStatus": "Active",
"applicationId": "WRS1274S1",
"applicationOfConformityLocationDetails": {
  "whLocationId": 19,
  "warehouseId": "WAR20250227001",
  "warehouseName": "Warehouse One",
  "whOperatorName": "Gyana",
  "email": "gyana@gmail.com",
  "mobileNumber": "0786567867",
  "lrNumber": "LR876567876",
  "location": "Nairobi",
  "longitude": "37.96234",
  "latitude": "-1.8745678",
  "streetName": "Street One",
  "building": "Demo Building",
  "policyNumber": "POLAA654678",
  "company": {
    "profDetId": "COP100041",
    "applicantName": "Gyanaranjan",
    "mobileNumber": "0798767689",
    "email": "gyana.testing@email.com",
    "postalAddress": "P.O. Box 56978-09089",
    "postalCode": "00800",
    "town": "Nairobi",
    "telephone": "+254700878999",
    "website": "https://www.google.com",
    "typeOfEntity": "Company",
    "company": "Gyana Company",
    "companyRegNo": "CRC231233453",
```

```
        "kraPin": "CV123123DC",
      },
    },
    "internalUserId": 2,
    "govtIssuedId": "1",
    "vchIdNo": "455555"
  },
  "commodityType": 1,
  "commodityOrigin": "North East",
  "commodityCode": "3223488",
  "originalQuantity": 345,
  "originalGrossWeight": 34,
  "originalNetWeight": 234,
  "currentQuantity": 45,
  "currentNetWeight": 234,
  "cropYear": {
    "seasonname": "Summer",
    "id": 3
  },
  "grade": "23",
  "lotNumber": "34",
  "qualityInspectionPath": "5731747718484785.pdf",
  "weighingTicketPath": "5981747718490775.pdf",
  "grnPath": "2151747718497792.pdf",
  "selectDate": "",
  "issuaneWarehouseOperatorCharges": [
    {
      "intIssueOperatorId": 11,
      "chargeHeaders": {
        "intChargesHeader": 3,
        "vchChargesHeader": "Re-bagging charge",
        "intUnitType": {
          "intUnitType": 3,
          "vchUnitType": "amount per bag or ton",
        }
      },
      "unitCharge": 234,
      "quantity": 345,
      "totalCharge": 80730,
      "grandTotal": 80730
    }
  ],
  "tradeStatus": false,
  "pledgeStatus": true,
  "pledge": [
    {
      "financialInstitution": "bank1",
      "dateOfPledging": "2025-05-20",
      "proposedLoanAmount": 95000,
```

```
        "proposedDateOfLoanClearance": "2026-03-12",
        "loanStatus": "Sanctioned",
        "durationOfLoan": 9,
      }
    ],
    "receiptStatus": "Issued",
    "wareHouseReceiptNo": "WR-124-2025",
    "paymentStatus": "Received",
    "vchWarehouseId": "WAR20250227001",
  }
}
```

Status Codes:

- 200 - OK
- 400 - Bad Request
- 401– Unauthorized (Invalid token)
- 404– Not Found (No data for provided ID)
- 500– Internal Server Error (Third-party service failure)